

Acoustic Emission Characterization of Damage in Hybrid Fiber-Reinforced Polymer Rods

Yujin Liang¹; Changsen Sun²; and Farhad Ansari³

Abstract: The development and testing of a pseudoductile hybrid fiber-reinforced polymer rod consisting of glass and carbon fibers is described. Two different hybrid forms were considered for the development of the hybrid rods, including (1) random dispersion of the glass and carbon fibers; and (2) a core-shell design with a glass fiber core protected by the carbon fiber shells. The mechanical properties of the hybrids were evaluated through uniaxial tension tests. The hybrid rods developed by way of the core-shell approach exhibited pseudoductile properties that included a yield point of 1,153 MPa and an ultimate strength of 1,191 MPa, with a strain of 3.5% at ultimate. The experimental program also included damage assessment by way of a single-channel fiber optic acoustic emission (AE) sensor. The acoustic emission technique was employed for real-time determination of the progressive damage due to rupture of fibers. Moreover, the spectral energies of the frequency spectrum from the AE signals were employed for assessment of the state of damage in the rods. The spectral energies distinguish between the carbon and glass fiber ruptures. This differentiation of signals provides information about the condition and the state of health of the hybrid rod.

DOI: 10.1061/(ASCE)1090-0268(2004)8:1(70)

CE Database subject headings: Rods; Fiber-reinforced polymers; Damage; Sensors; Glass fibers; Acoustic detection.

Introduction

The structural capacity and serviceability of prestressed and post-tensioned structures using fiber-reinforced polymer (FRP) composites as tendons have been the topic of many research programs, including the work of Dolan (1990), Naaman (1993), Abdelrahman and Rizkalla (1997) and Grace and Sayed (1998). These composites exhibit higher strength than steels; however, their stress-strain behavior is linear elastic all the way to failure (Fig. 1). This is in contrast to the elastoplastic and ductile behavior of steel. Consequently, the energy release of beams with FRP tendons is mainly elastic and sudden, which is different from the inelastic energy consumed prior to failure of beams prestressed by steel tendons. An area of research for improving the ductility of FRP rods pertains to development of hybrid rods by way of combining two or more materials in order to achieve ductile or pseudoductile characteristics (Fig. 2).

Recent research in hybrid FRP composites for civil structural application dates back to the 1990s, including the works of Phelan and Triantafillou (1992), Nanni et al. (1994), Bakis et al. (1996, 2001), Belarbi et al. (1996), Tamuzs et al. (1996), and Somboonsong et al. (1998). Phelan and Triantafillou (1992) presented a design approach for producing hybrid rods exhibiting bilinear

stress-strain behavior through combination of glass and carbon fibers. The glass fibers were insulated from the concrete by carbon fibers and the epoxy matrix. Nanni et al. (1994) developed and tested bars of braided Aramid fiber around a steel core in an epoxy matrix and obtained bilinear stress-strain behavior. They concluded that the rule of mixtures could be used for predicting elastoplastic behavior. Bakis et al. (1996) reported results from pultruded hybrid FRP bars composed of different types of glass, Aramid, carbon, and vinylon fibers. One of the major findings of their research pertained to maximization of “pseudoductile” behavior by thorough dispersion of higher-modulus fibers over the entire cross-sectional area. Bakis et al. (2001) took advantage of the piezoresistive properties of carbon fibers and developed an effective self-monitoring methodology for characterization of deformation and damage throughout the rods. Tamuzs et al. (1996) recognized the advantage of braided structures in increasing the ductility of FRP by optimization of braid angle with respect to the core structure. Belarbi et al. (1996) theorized that a hybrid composite consisting of several brittle fibers of different stiffness and ultimate strains could be manufactured to behave as a pseudoductile material. These hybrids were tested under uniaxial tension and in reinforced concrete beams. Somboonsong et al. (1998) developed a hybrid rod that possessed a definite yield point and achieved ultimate strains in the 2–3% range.

Failure of FRP rods is accompanied by breakage of individual fibers and rapid localization of damage, leading to sudden brittle failure. Properly designed hybrid bars will delay brittle failure by allowing progressive breakage of fibers and gradual redistribution of stresses from the broken to the undamaged strands. The objective of the present study was to develop a composite rod with elasto-pseudoplastic stress-strain characteristics in order to allow for stress redistribution during ultimate stages of failure. It is understood that the current state of the art in composites as it pertains to pseudoductile materials is still at the stage of infancy. At this point in time, even with successful design and production of pseudoductile materials, their ultimate strain capacities will not

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Note. Discussion open until July 1, 2004. Separate discussions must be submitted for individual papers. To extend the closing date by one month, a written request must be filed with the ASCE Managing Editor. The manuscript for this paper was submitted for review and possible publication on June 4, 2002; approved on September 20, 2002. This paper is part of the *Journal of Composites for Construction*, Vol. 8, No. 1, February 1, 2004. ©ASCE, ISSN 1090-0268/2004/1-70-78/\$18.00.

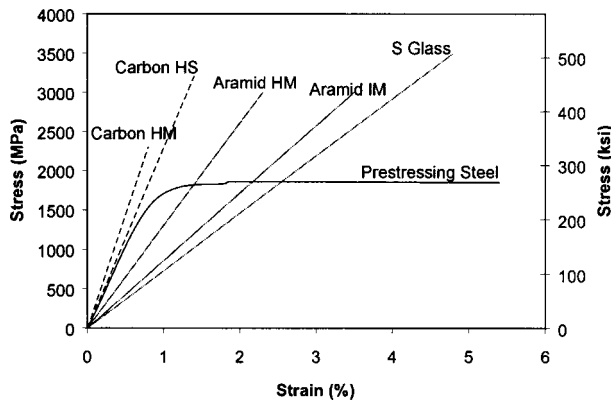


Fig. 1. Comparison of stress-strain characteristics of various types of reinforcing fibers

approach those enjoyed by steels. While the added strain capacity in the pseudoplastic stage enhances the ductility of the beam, the increase in strain capacity is not sufficient to carry sustained levels of overstress. It will be necessary to develop customized condition monitoring technologies for assessment of tendon integrity, especially as it pertains to identification of the elastic and pseudoplastic states of failure. Since in most of the structural elements the tendons are encased in ducts, it becomes necessary to employ nondestructive evaluation (NDE) techniques. Hence, the goal of the present work was twofold: (1) to design and manufacture a pseudoductile hybrid FRP rod and evaluate its mechanical properties through uniaxial tensile tests; and (2) to evaluate the feasibility of using specific features of the acoustic emission (AE) for assessing the state of damage in the hybrid rod.

A primary goal of the research was to achieve near real-time monitoring capability in terms of acquisition, analysis, and display of the results. For this reason, the damage assessment methodology presented here is accomplished by way of a fiber optic based acoustic emission sensor. Standard specifications for proper testing of FRP components such as vessels, tanks, and pipes by way of acoustic emissions have existed for quite some time (ASTM 1998). Acoustic emissions are transient mechanical vibrations generated by rapid release of energy from localized sources within materials. The main source of AE generation in FRP composites pertains to fiber breakage, although, depending on the geometry, size, loading condition, and morphology of the FRP component, AE's may be generated by other sources, including matrix cracking, fiber pullout, and laminate separation. Some recent activity in the applications of AE techniques in civil engi-

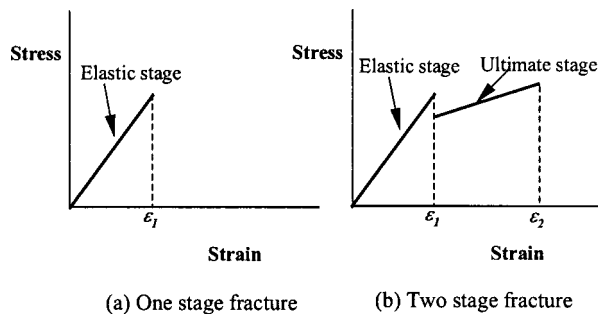


Fig. 2. Idealized elastic and pseudoductile stress-strain curves: ϵ_1 = ultimate strain for one-stage fracture, ϵ_2 = ultimate strain for two-stage fracture

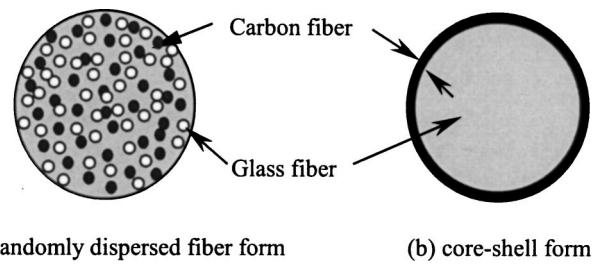


Fig. 3. Morphology of hybrid forms: (a) randomly dispersed fiber form; (b) core-shell form

neering includes the work of Chen et al. (1993), Maji and Shah (1994), Maji et al. (1997), and Mirmiran et al. (1999). Chen et al. (1993) employed the AE technique for locating the damage points along the length of AFRP (Aramid FRP) rods. Their work included tensile testing of AFRP and use of the AE signals to detect the location of fiber rupture points. Maji and Shah (1994) developed source location methodologies in reinforced concrete, and later Maji et al. (1997) employed Lamb waves for characterization and detection of damage. Mirmiran et al. (1999) employed the AE technique for condition monitoring of hybrid concrete-FRP beam columns consisting of concrete-filled FRP tubes. Their work included cyclic testing of specimens through which he was able to establish a definitive correlation between the AE activity and the extent of damage in FRP-concrete systems. The fiber optic AE technique described here is used for real-time damage assessment of the hybrid rods.

Proportioning of Hybrid Fiber-Reinforced Polymer Rods

The hybrid rod described in the present study was proportioned by combining carbon and S-glass fibers within the epoxy matrix. The strength and stiffness of the hybrid were derived from the carbon fibers, whereas the glass fibers provided the added strain capacity. Two basic hybrid forms were considered: (1) random dispersion of the two fiber types across the cross section; and (2) a core-shell form in which the carbon fibers provided the outer shell for the glass core. The latter arrangement also protects the glass fibers from the alkaline environment in cementitious materials (Fig. 3). The rule of mixtures was used for proportioning of the fibers in the hybrid. For simplicity, a perfect interfacial bond is assumed between the fiber and the matrix and the tensile modulus of the epoxy matrix is ignored ($E_m \ll E_c$ or E_g). The modulus of elasticity of the hybrid considering arbitrary proportions of carbon and glass fibers is computed as

$$E_H = (1 - V_m)(E_c V_c + E_g V_g) \quad (1)$$

where E_H = tensile modulus of the hybrid FRP rod; E_c and E_g = tensile moduli of carbon and glass fibers, respectively; V_m , V_c , and V_g = volume fractions of the matrix, carbon, and glass fibers, respectively. The strength of the hybrid at the elastic and ultimate stages of rupture are computed

$$\sigma_{H,1} = (1 - V_m)\epsilon_c(E_c V_c + E_g V_g) \quad (2)$$

$$\sigma_{H,2} = (1 - V_m)\epsilon_g E_g V_g$$

where $\sigma_{H,1}$ and $\sigma_{H,2}$ = tensile strengths of the hybrid rod at the elastic and ultimate stages of rupture, respectively; whereas ϵ_c and ϵ_g = ultimate strains of carbon and glass fibers, respectively.

Table 1. Properties of Hybrid Fibers and Matrix

Property	S2 glass fiber	Carbon fiber	Epoxy matrix
Tensile modulus [MPa (ksi)]	89,635 (13,000)	230,293 (33,400)	Ignored
Tensile strength [MPa (ksi)]	4,275 (620)	4,930 (715)	Ignored
Elongation (%)	5.0	2.1	Ignored
Volume percent in rod (%)	48	13	39

Uniaxial straining of the rod subjects the glass and carbon fibers to different levels of stress. The carbon fibers rupture first since their ultimate strain capacity is lower than the strain capacity of the glass fibers. Depending on the volume percentages of these two fibers, there will be either a one-stage brittle fracture of the rod, or a two-stage elastoplastic failure as shown in Fig. 2. To achieve pseudoductile failure, the volume of glass fibers should exceed a certain threshold to assure smooth redistribution of stresses from the ruptured carbon fibers. This can be achieved by setting $\sigma_{H,1} = \sigma_{H,2}$, from which the threshold volume of glass fiber is derived

$$V_g = \frac{\varepsilon_c E_c}{(\varepsilon_g - \varepsilon_c) E_g} V_c = V_{cr} \quad (3)$$

Accordingly, when $V_g > V_{cr}$, the rod is elastoplastic, otherwise the failure will be brittle.

To evaluate the above approach, the research program involved evaluation of the mechanical properties of 9.5 mm (0.375 in.) diameter pultruded hybrid rods. Both randomly dispersed and core-shell hybrids were considered. The volume percentages of all the constituents in the hybrid rod were computed accordingly to achieve pseudoductile behavior. Table 1 lists the mechanical properties and the volume percentages of the constituents. Encasement of fibers within the matrix during the pultrusion process introduces residual stresses in the rod. This tends to lower the overall strength of the hybrid. The volume percentages of the constituents were consequently adjusted to compensate for this strength loss.

Experimental Program

The experimental program included uniaxial tensile tests of the instrumented hybrid rods in a servohydraulic closed-loop materials testing system with a load capacity of 667.2 kN (150 kips). Specimen designations and types are given in Table 2. All the rods possessed a nominal diameter of 9.5 mm (0.375 in.). Bond type anchorages were used for gripping the rods during the tension tests. They were manufactured by machining steel tubes to a specific length for development of bond between the hybrid rod and the cement inside the anchorage system (Gu and Ansari 2000). Expansive cement was employed for the development of the bond in the anchorages. The full bond between the carbon-glass FRP (CGFRP) rod and the cement was developed after six days of curing at 21°C (70°F). Anchorage details and dimensions are shown in Fig. 4. The 63.5 cm (25 in.) nominal distance between the two anchorages in each specimen corresponded to the gauge length for the linear variable displacement transformer (LVDT). The LVDT with a displacement range of ± 12.7 mm (± 0.5 in.) was employed for measuring the axial deformation. Four strain gauges were glued at the midheight of the hybrid rod,

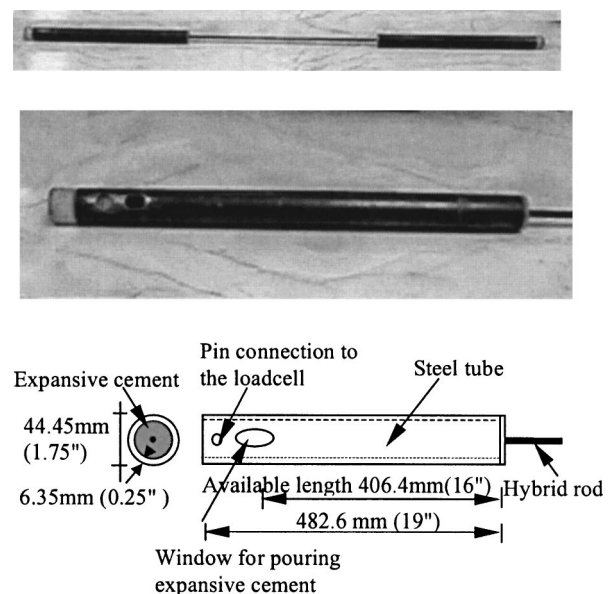
Table 2. Specimen Designations

Specimen designation	Specimen type	Instrumentation	Acoustic emission tests
CGFRP-DF1	Randomly dispersed fibers	LVDT and four strain gauges	No
CGFRP-DF2	Randomly dispersed fibers	LVDT and four strain gauges	No
CGFRP-DF3	Randomly dispersed fibers	LVDT and four strain gauges	No
CGFRP-CS4	Core-shell	LVDT and four strain gauges	No
CGFRP-CS5	Core-shell	LVDT and four strain gauges	No
CGFRP-CS6	Core-shell	LVDT and four strain gauges	PZT and fiber optics
CGFRP-CS7	Core-shell	LVDT and four strain gauges	PZT and fiber optics
CGFRP-CS8	Core-shell	LVDT and four strain gauges	PZT and fiber optics

two for measurement of axial strains (for cross checking the LVDT results) and two for transverse strains. Two computers were used, one for data acquisition and the other for providing control commands to the servohydraulic testing system. The specimens were loaded in load control up to about 60% of the computed yield strength, and then the control was transferred to displacement control for the duration of the experiment. A typical tension test setup in the servohydraulic system and the schematic detailing the instrumentation for the tensile test are shown in Fig. 5.

Acoustic Emission Monitoring

Optical fibers have been employed for sensing of acoustics in naval and aeronautics applications. Stress waves generated in materials by way of acoustic emissions perturb the light propagating through the optical fiber (Fig. 6). Interrogation of this perturbed

**Fig. 4.** Anchorage system for hybrid fiber-reinforced polymer rod

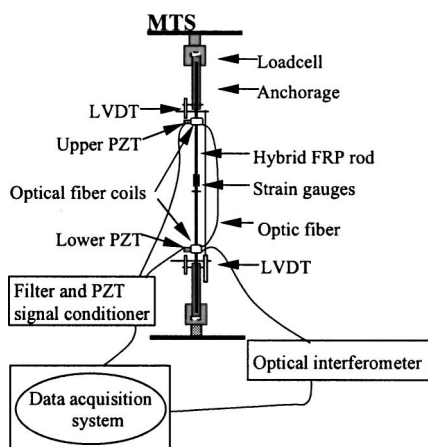
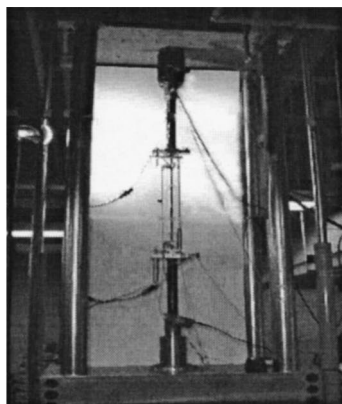


Fig. 5. Test setup for hybrid fiber-reinforced polymer rods

light is used for nondestructive evaluation of the structural element. In general, this is done interferometrically by measuring the AE generated phase delays within a gauge length of the optical fiber. Giallorenzi et al. (1982) detailed the basic principles and methodologies involved in fiber optic sensing of stress waves and acoustical signals. There are certain advantages associated with fiber optic sensing of acoustic emissions, among which is the capability for making broadband measurements with a great degree of sensitivity. This feature facilitates differentiation of the damage source among the various AE events in heterogeneous materials by way of frequency domain analysis methods. Liu et al. (1990) employed a fiber optic acoustic sensor for detection of flaws in Kevlar FRP composites. They were able to detect the air voids in the epoxy during the material processing stage of the composite production. Pierce et al. (1996) performed fundamental

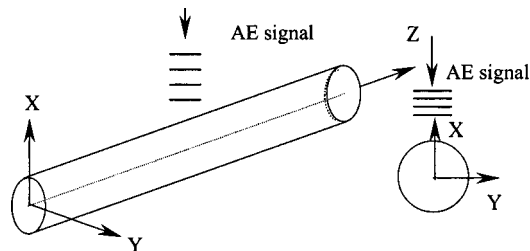


Fig. 6. Interaction between optical fiber and acoustic emission signals

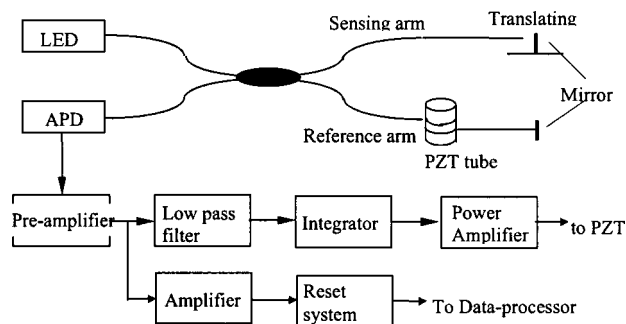


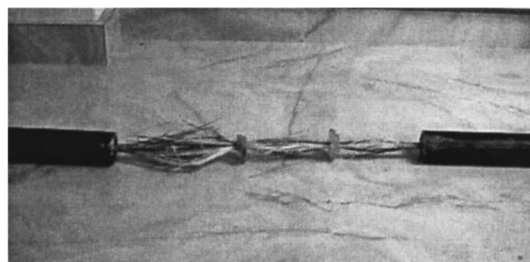
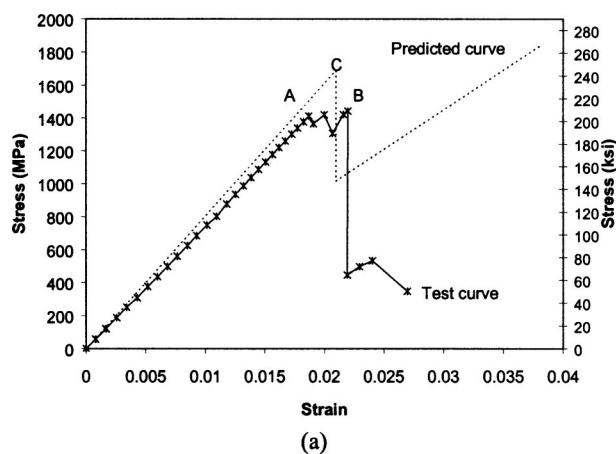
Fig. 7. Fiber optic acoustic emission interferometer

experiments with a Michelson interferometric fiber optic sensor. Their work involved embedment as well as adhesion of the fiber optic sensor in carbon FRP composites. Their work demonstrated the superior sensitivity of optical fibers in detection of Lamb waves during splitting of the fibers. Chen and Ansari (2000) developed a novel optical fiber sensor with the capability of making distributed measurements. The design of the sensor involved development of AE sensing regions along the length of an optical fiber. A white light Michelson interferometer was used for interrogation of the AE events at their quadrature points to achieve maximum sensitivity. Chen and Ansari's work demonstrated that a single optical fiber with two sensitized regions is capable of locating the source of an AE event in a rodlike element.

The single-sensor arrangement is advantageous for real-time nondestructive evaluation of structures by simplifying the instrumentation through single-channel data acquisition, signal analysis, and real-time conversion of the AE arrival times to location of the AE source. An interferometer similar to the one described by Chen and Ansari (2000) was employed in the present study (Fig. 7). Details pertaining to the design and operational principles of the system including development of sensing regions along the length of the optical fiber are given elsewhere (Chen and Ansari 2000). A brief description is given here for completeness. In Fig. 7, LED and APD are the light emitting diode and avalanche photo diode for transmission and detection of optical signals, respectively. The function for the piezoelectric transducer (PZT) is to resonate the interferometer at the working point of the fringe patterns to attain high-resolution time domain signals. The sensing arm of the sensor is sensitized at two locations to create two high-resolution sensing regions along the length of the fiber. Sensitized regions are created by wrapping the fibers into coils with precise diameters to achieve the frequency range of interest. The two sensitized segments were constructed 63.5 cm (25 in.) apart for connection at the two gauge-length ends of the CGFRP rods. As shown in Fig. 5, two PZT sensors were also attached to the CGFRP rods for comparison of AE arrival times with those obtained from the fiber optic sensor.

Stress-Strain Response of Hybrid Rods

The first series of tensile tests was performed on CGFRP rods with randomly dispersed fibers. The stress-strain response and the failure characteristics of the randomly dispersed fiber rods were different from the predicted behavior. Point A in Fig. 8(a) corresponds to rupturing of the carbon fibers at a tensile stress of 1,440 MPa (209 ksi) and a strain of 1.87%, whereas point C denotes the predicted value computed from Eq. (2). Following the rupturing of the carbon fibers, the glass fibers began to fracture rapidly, yet not



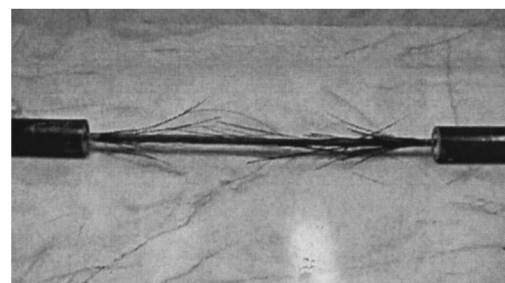
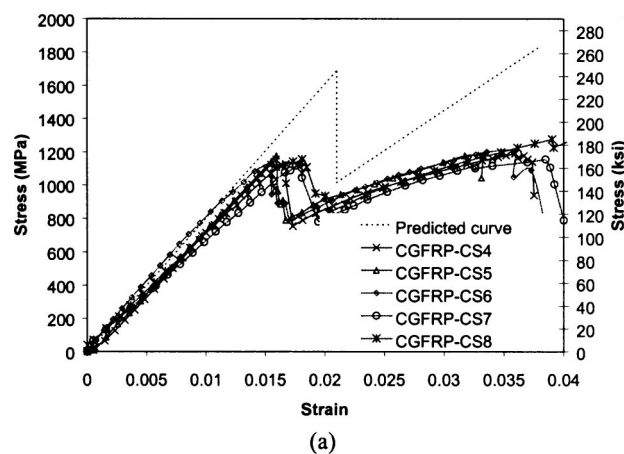
(b)

Fig. 8. Typical stress-strain response and failure characteristics of randomly dispersed fiber hybrid rods

in a sudden brittle manner. The strain gauges ceased functioning once the fracturing of the fibers commenced. Axial strains were acquired from the measurements made by the LVDT. The hybrid rod exhibited what could be considered a pseudoductile behavior as shown by the trend between points A and B in Fig. 8(a), followed by a sudden drop in load to failure. The failure was not predictable and the specimens were totally shattered by the completion of the experiments [Fig. 8(b)]. A number of factors may have contributed to the undesirable behavior of the randomly dispersed fiber hybrid design:

1. The random dispersion of fibers did not allow for smooth redistribution of the stress from carbon to glass fibers.
2. The energy release due to individual fracturing of carbon fiber strands caused premature rupture of the adjacent glass fibers leaving fewer of these fibers for stress redistribution ($\leq 48\%$).

The second series of experiments involved CGFRP rods of core-shell design. As shown in Fig. 9(a), the stress-strain response starts with a linear elastic stage and then exhibits a sudden change in stiffness characterized by a sharp “knee” in the curve with a drop in stress, where the stress is again increased to an ultimate value higher than the pseudoyield point of the material. Acoustic emission results indicated that the carbon fibers began fracturing at stresses slightly larger than 70% of the yield (Fig. 10), and the yield point pertained to the rupture of the greater part of fibers in the carbon shell at an average stress level of 1,153 MPa (167 ksi) corresponding to a strain of 1.57%. The remaining carbon fibers fractured along with some of the glass fibers during the postyield unloading phase, in the segment between points B and C in Fig. 10. As will be shown in the following section, AE events made it possible to differentiate between the fiber type ruptures, i.e., carbon or glass fibers. The stress drop following the yield was steep but gradual until the glass fibers fully took over to carry the



(b)

Fig. 9. Stress-strain response and failure characteristics of core-shell hybrid rods

additional strain of 2% prior to failure at an average ultimate stress of 1,191 MPa (173 ksi) and a strain of 3.5%. Fracturing of the carbon fibers did not induce as much damage in the glass fibers as previously observed with the dispersed fiber specimens [Figs. 8(b) and 9(b)]. An average Poisson's ratio of 0.285, and a modulus of elasticity of 72,620 MPa (10,532 ksi) was computed for the hybrid rods based on the measured values of transverse and axial strains during the elastic phase of the loading.

Damage Assessment

Typical acoustic emission event counts acquired during a tension test are shown in Fig. 10. As indicated in this figure, AE events

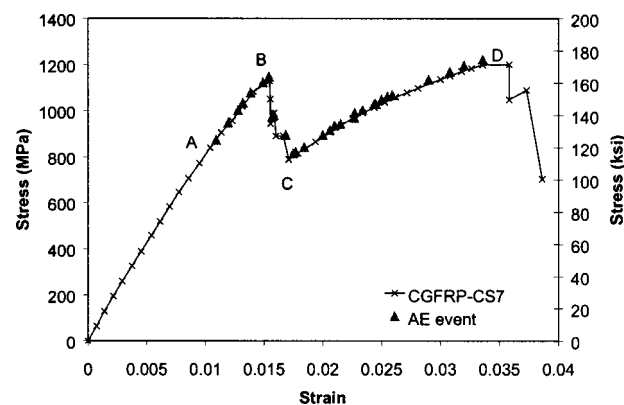


Fig. 10. Typical acoustic emission event counts during tensile tests (CGFRP-CS7)

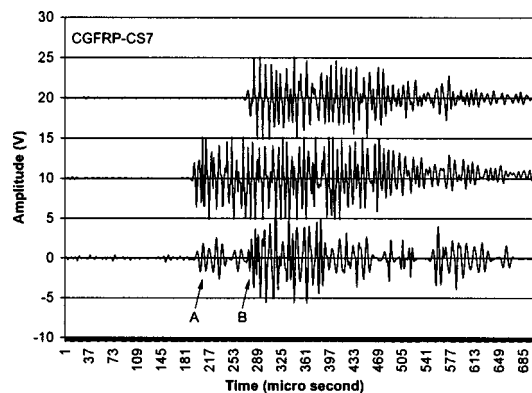


Fig. 11. Typical acoustic emission signals detected by optic fiber and two piezoelectric transducers

commenced at about 70% of the yield point and continued on through the ultimate stage until failure. The AE events provided two types of information: (1) location of the damage; and (2) state of the damage, i.e., whether the damage occurred prior to the yield point (rupture of carbon fibers) or afterward during the ultimate stage (rupture of glass fibers).

Time domain signals from a typical AE event are shown in Fig. 11. The two upper signals in Fig. 11 (trace I and trace II) correspond to the AE event detected by the upper and lower PZTs. The third signal is the same AE event captured by the optical fiber. Since the AE event is captured at two different locations along the same fiber, both signals are captured on the same channel. Points A and B in Fig. 11 (trace III) pertain to the arrival times of the pulse at the upper and lower portions of the rod. The time difference in arrival times is instantaneously related to the location of the AE event. The linear source location method was employed for this purpose (Chen et al. 1993). This technique employs the time of arrival of an AE event at two different locations according to the following relationship:

$$\text{location} = \frac{T_{\max} - \Delta T}{2} \times \frac{\text{gauge length}}{T_{\max}} \quad (4)$$

where $T_{\max}$ = travel time between the two sensing points, which is inversely proportional to the speed of the stress wave in the rod material; and ΔT = time of arrival difference for the AE signal sensed at the two sensing points in the optical fiber. $T_{\max}$ was measured during preliminary experiments by transmission of a pulse from one end of the hybrid rod and capturing the signal from the other end. The values of $T_{\max}$ measured at the elastic stage, where both carbon and glass fibers were intact within the rod, and at the ultimate stage (ruptured carbon fibers) were 117.0 and 99.2 μs , respectively.

Fig. 12 depicts the AE detected locations of the fiber ruptures along the length of the hybrid rod during a typical tension test. The AE source locations were also computed from the time domain signals acquired by the two PZT transducers. As shown in Fig. 13, the error between the two measurements is larger for the AE signals that occurred within the middle section of the rod. The sensors detect the AE signals generated from the midsection of the rod almost synchronously, and therefore a high demand is placed on the temporal resolution of both sensor types.

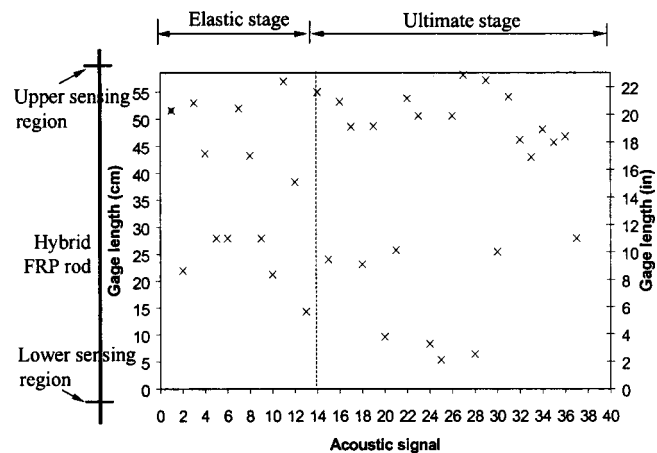


Fig. 12. Location of fiber rupture points in hybrid rod (CGFRP-CS8)

Damage Identification

Time domain signals provide information about the source of damage, but they are in general very complicated for use as a damage assessment tool. An alternative approach is to use frequency analysis, especially since the optical fiber contains more information than the individual PZT transducers. Frequency spectra of individual AE events at various stages of loading were obtained through fast Fourier transform analysis of time domain data. Representative frequency spectrum plots of the AE events are shown in Fig. 14, the plots for the damage events in the elastic regime are shown in the left column [Fig. 14(a)], and the right column contains the spectral plots of damage events associated with the ultimate stage of loading [Fig. 14(b)]. Here, damage at the elastic stage pertains to the fracturing of the carbon fibers, and the ultimate stage damage represents fracturing of the glass fibers during the postyield phase of the loading. In both cases, the maximum peak within the frequency spectrum occurs somewhere within the range 100–120 kHz. However, comparison of Figs. 14(a and b) reveals that the signals emanating from the fracturing of the carbon fibers contain more of the lower-frequency, higher-amplitude signals than the signals stemming from the splitting of the glass fibers. This observation can be further justified through

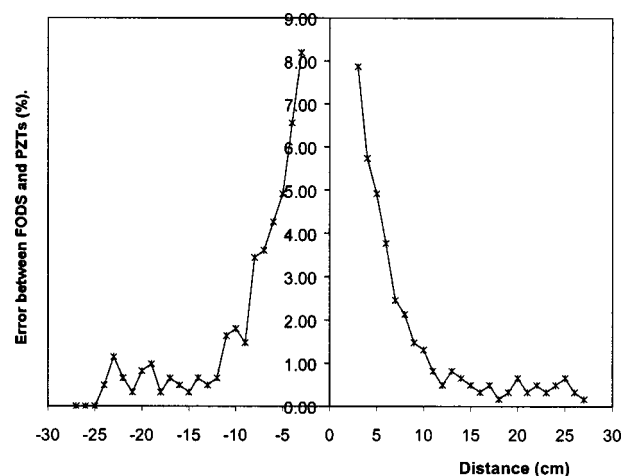


Fig. 13. Comparison of the piezoelectric transducer and fiber optic source location measurements along rod

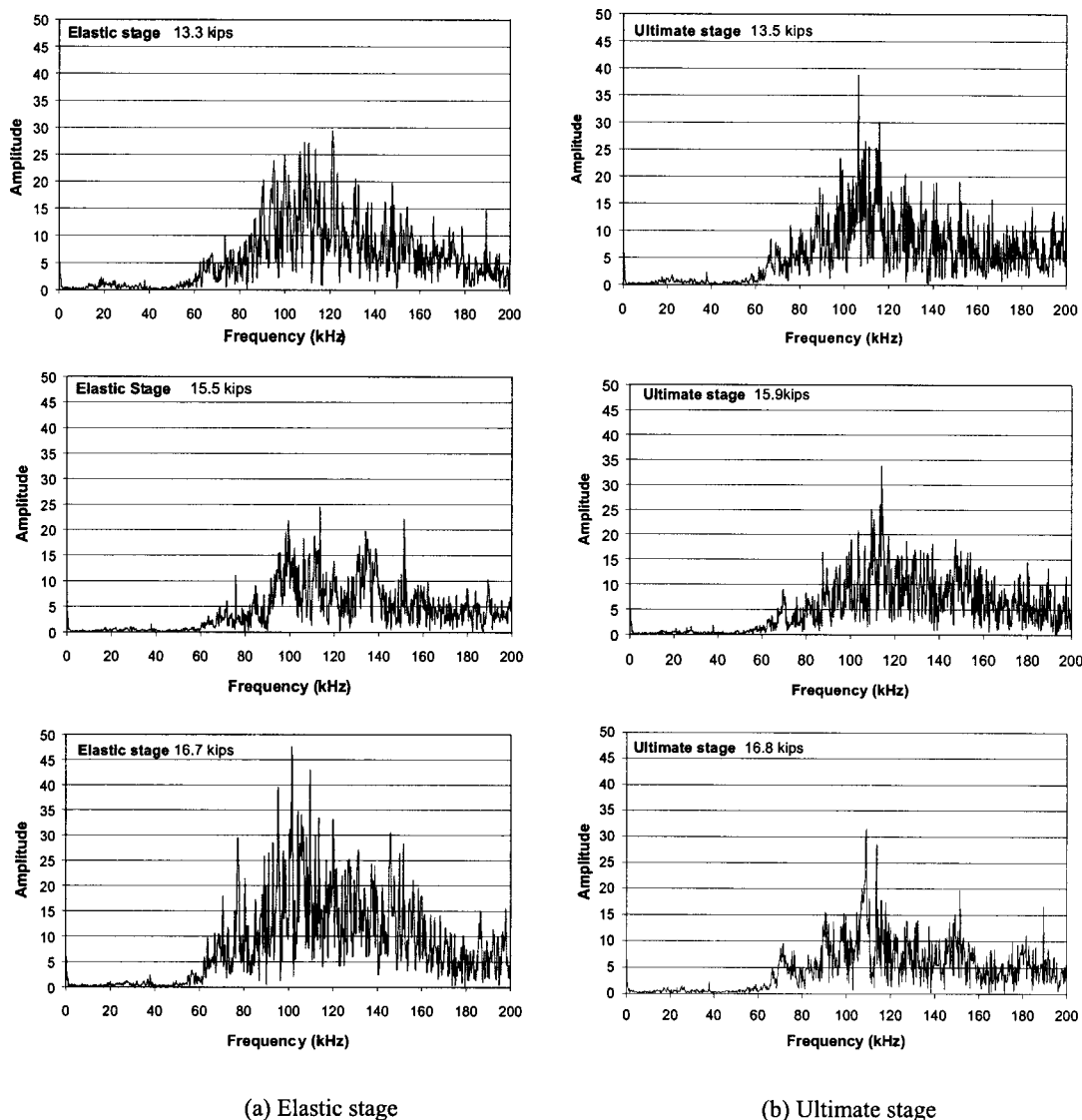


Fig. 14. Frequency spectrum plots of acoustic emission events: (a) Elastic stage; (b) ultimate stage

computation of the spectral energy of individual events. The normalized average spectral energy of AE events is computed from the following relationship:

$$E = \frac{\int_{\omega_i}^{\omega_p} |A(\omega)|^2 d\omega}{2\pi(\omega_p - \omega_i)A(\omega_p)^2} \quad (5)$$

where E = normalized average spectral energy; $A(\omega)$ = frequency amplitude; and ω_i and ω_p correspond to the initial and peak frequencies. Fig. 15 represents a plot of the spectral energies of individual AE events. As shown, except for a few points, the spectral energies of the elastic stage events are larger than the spectral energies of the AE events that occur during the ultimate stage. Alternatively, Fig. 16 represents the cumulative plot of the spectral energy as a function of the AE events. The spectral energy exhibits a bilinear relationship separating the elastic and ultimate states of damage. A cumulative plot of the spectral energy provides deterministic information about the change in the state of the hybrid rod. The spectral energy of an AE event is a function of the spectral signature of the individual fiber as well as the geometry and the state of health of the surrounding composite. As a result, this technique is most efficient for NDE and

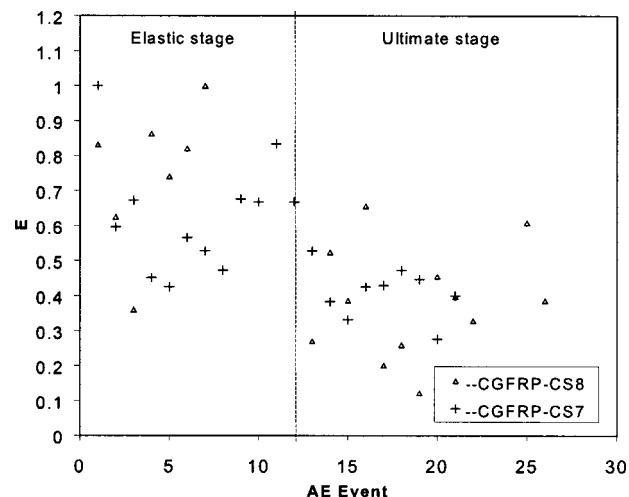


Fig. 15. Spectral energy plot of acoustic emission signals

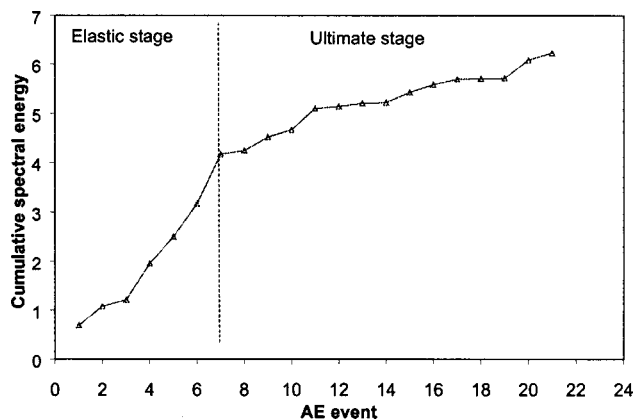


Fig. 16. Cumulative energy power spectrum

damage assessment in composites composed of more than one type of fiber, such as the case in the present study.

Conclusions

The design and development of a pseudoductile hybrid FRP rod consisting of glass and carbon fibers have been described. Two different hybrid forms, namely, randomly distributed and core-shell designs, were considered. The experimental results indicated the suitability of the core-shell approach for manufacturing the hybrid rods. The mechanical characteristics of the rods were evaluated through tensile tests. The results indicated enhanced ductility and strain capacity for the carbon-glass hybrid composite. However, the mechanical properties of the hybrid will still need further improvements prior to practical applications. There is an average drop of 337.3 MPa (49.3 ksi) in stress during the transition phase following the fracture of the carbon fibers and prior to the redistribution of stresses to the glass fibers. Further research is necessary to improve the mechanical characteristics of the material. For instance, one way to decrease the amount of the postyield stress drop during the transition stage is to increase the percent volume of glass fibers within the hybrid. Adjustments to the fiber type volumes need to be optimized in order to attain sufficient stiffness in the hybrid, which is derived from the carbon fibers.

The fiber optic single-channel acoustic emission technique provides the capability for real-time assessment of damage in the hybrid rods. In the present experiments, the damage location identification was done instantaneously during the course of the experiments. The damage phase identification process was performed through automated processing routines and it involved a time delay. The spectral energy approach is very effective for hybrid composites, as the signature frequency spectra of the individual fibers are used for distinguishing the state of damage. For single-fiber composites, the spectral energy will be solely a function of the geometry and state of health of the composite. Further research is needed to enhance the technology for such applications.

Acknowledgments

Financial support under Grant No. CMS-9900338 by the National Science Foundation is greatly appreciated. The hybrid FRP rods were fabricated by the DFI Pultruded Composites, Inc. in Erlanger, Ky.

Notation

The following symbols are used in this paper:

- $A(\omega)$ = frequency amplitude;
- E = normalized average spectral energy;
- E_c = tensile modulus of carbon fibers;
- E_g = tensile modulus of glass fiber;
- E_H = tensile modulus of hybrid FRP;
- $T_{\max}$ = travel time between two sensing points;
- V_c = volume fraction of carbon fiber;
- V_{cr} = threshold volume of glass fiber;
- V_g = volume fraction of glass fiber;
- V_m = volume fraction of matrix;
- ΔT = time of arrival difference for AE signal;
- ε_c = ultimate strain of carbon fibers;
- ε_g = ultimate strain of glass fibers;
- $\sigma_{H,1}$ = tensile strength of hybrid rod in elastic stage;
- $\sigma_{H,2}$ = tensile strength of hybrid rod in ultimate stage;
- ω_i = initial frequency; and
- ω_p = peak frequency.

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